INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, MB-4

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description (CID) covers the requirements for the MB-4 aircraft towing tractor, hereafter referred to as tow tractor. The tow tractor is capable of towing munitions trailers weighing up to 80,000 pounds and military aircraft weighing up to 175,000 pounds on dry level concrete on U.S. Air Force installations.

2. CLASSIFICATION.

2.1 Tow tractor classes. The two classes specified shall be of the same design, with the only differences being in the components necessary for an air system for towing the Air Force's four-wheel MHU-196 and MHU-204 munitions trailers with air brakes. The tow tractor shall be one of the following classes specified in Table I:

TABLE I. Classification.

Class 1	tow tractor without air system for munitions trailer towing (standard equipment)
Class 2	tow tractor with air system for munitions trailer towing (optional equipment, see 7.3)

3. SALIENT CHARACTERISTICS.

3.1 MB-4 description. The class 2 MB-4 tows MHU-196 and MHU-204 munitions trailers in support of bomber aircraft and is able to tow the smaller MHU-110, MHU-141, and MHU-226 munitions trailers in support of other weapons-delivering aircraft. Both classes of the MB-4 may

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tow any trailers that have a lunette-towing hitch and fall within the towing requirements of the MB-4. The MB-4 tows C-130 and other medium-sized aircraft and is used to tow fighter aircraft.

3.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. The tow tractor shall be designed and manufactured in accordance with Society of Automotive Engineers (SAE) specifications ARP1247 and AIR1363. The tow tractor shall be designed and constructed so that no parts will work loose in service, and it shall be designed and constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The tow tractor shall have an all-welded, one-piece frame structure.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used.

3.2.1.2 Finish. The exterior finish color of the tow tractor shall be the tow tractor manufacturer's standard commercial color. When specified (see 7.3), the exterior finish color shall be gloss white. Prior to the application of the finish top coat, any cab or chassis component or assembly specifically manufactured by or for the tow tractor manufacturer, from sheet, plate, angular, or tubular steel, shall be coated with a zinc-rich primer. The top surfaces of the tow tractor that are walked over in association with routine maintenance shall be covered with anti-skid material.

3.2.1.3 Exclusion of water. The design of the tow tractor shall be such as to prevent water leaking into, or being driven into, any part of the tow tractor interior when either in an operating or travelling configuration. All windows, doors, panels, covers, or related type items shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the tow tractor that could collect and retain water.

3.2.1.3.1 Fluid traps and faying surfaces. There shall be no fluid traps on the tow tractor. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.3.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.3.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to affect maximum drainage of accumulated fluids. The number and location of drain holes shall permit drainage of all fluids. The minimum size of the drain holes shall be 0.25 inch.

3.2.1.4 Corrosion protection.

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3.2.1.4.1 Permanent corrosion protection. The tow tractor shall be corrosion-protected with a permanent, black, commercially available Corrosion Preventive Compound (CPC) that is difficult to remove (Cortec VpCI 375 or equivalent). Application shall include the exposed chassis-frame metal underneath the tow tractor (excluding the drive train), the inside of the wheel wells, and behind the fender skirts. Any equivalents used shall meet the below salient characteristics in table II when applied on AISI carbon steel. In addition to the equivalent requirements in table II, the pencil hardness in accordance with American Society for Testing and Materials (ASTM) D3363 shall be "H," the temperature resistance shall be 150–350 °F, and the adhesion in accordance with ASTM D3359 shall be "5B."

TABLE II. VpCI 375 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	4.0-4.5	2500+
ASTM D1748	4.0-4.5	2000+
ASTM D870	4.0	312+

3.2.1.4.2 Temporary corrosion protection. When specified (see 7.3), tow tractors that are to be shipped overseas shall be temporarily corrosion-protected from the effect of salt spray both on the ship and at port with the application of a thin, soft-film, semi-transparent, commercially available CPC that is easily removable by washing or with a low pressure steam (Cortec VpCI 389 or equivalent). Application of the temporary CPC shall be to the exterior of the tow tractor and in critical or corrosion sensitive areas of the tow tractor. Any equivalents used shall meet the below salient characteristics in table II when applied on AISI 1010 carbon steel:

TABLE III. VpCI 389 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	2.0	600+
ASTM D1748	1.0	1000+
ASTM G85 (94)	1.0	600+

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1.00-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Black Int'l, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Red Int'l, Color Number 31136 of SAE AMS-STD-595. Each side of the tow tractor shall be marked with a center of gravity decal applied within 1.00 inch of the calculated center of gravity. Tire inflation pressures shall be marked above all wheel well openings, with 0.50-inch high letters/numbers that are black in color. A decal shall be adhered near the top of each wheel well opening that warns personnel of the dangers of placing any part of the body into the wheel well area. The warning shall be documented in the Technical Manual. A decal shall be included on the front and rear of the tow tractor near the coupler position that states: "MAXIMUM TOWING CAPACITY: 16,000 LBS DRAWBAR PULL FORCE / 80,000 LBS GVW TRAILERS / 175,000 LBS GVW AIRCRAFT." A decal shall be included on the rear of the class 2 tow tractor

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near the upper coupler position that states: “USE UPPER COUPLER POSITION TO TOW MHU-196/204 TRAILERS.” A decal shall be included on the front of the class 2 tow tractor between the coupler positions that states: “DO NOT USE THE FRONT COUPLER POSITIONS TO TOW MHU-196/204 TRAILERS.”

3.2.3 Identification and data plates.

3.2.3.1 Identification plate. The tow tractor shall be provided with a permanently-marked identification plate constructed of a non-corroding metal, mounted on the inside of the tow tractor cab, and a rear license plate mounting bracket with license plate attaching hardware. The variable information on the identification plate shall be engraved onto the plate by a method that produces a consistent lettering quality. The invariable information contained on the identification plates shall be documented in the Technical Manual. The tow tractor shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate, marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings. At a minimum, the identification plate shall include the following information:

a. Vehicle Nomenclature.

b. Manufacturer’s Part Number and/or Model Number.

c. Vehicle Identification (serial) Number or VIN.

d. Date of Manufacture.

e. Manufacturer’s name, address, and phone number.

f. Commercial and Government Entity (CAGE) code.

g. National Stock Number.

h. Contract Number.

i. Registration Number.

j. Date of warranty expiration.

3.2.3.2 Transportation data plate. The tow tractor shall be provided with a permanently marked transportation data plate constructed of a non-corroding metal, mounted on the exterior surface of the tow tractor. The information contained on the transportation data plate shall be documented in the Technical Manual. The plate shall contain at least the following information:

a. Side and rear silhouette views of the tow tractor.

b. Horizontal and vertical location of the center of gravity of the tow tractor in air transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

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3.2.3.3 Lubrication data plate. The tow tractor shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the tow tractor that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 3.3.1. The information contained on the lubrication data plate shall be documented in the Technical Manual.

3.2.4 Environment, Safety, and Occupational Health (ESOH).

3.2.4.1 Component protection. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.2 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the tow tractor with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.2.4.3 Fire extinguisher. The tow tractor shall be equipped with two 5 pound (minimum) dry chemical type fire extinguishers with mounting bracket suitable for use on vehicles and an Underwriters' Laboratories (UL) rating of 20B:C (minimum). The fire extinguishers shall be mounted on the left and right side of the tow tractor, rear of the cab.

3.2.4.4 Back-up alarm. An externally mounted back-up warning alarm shall be included in the reverse light circuit.

3.2.5 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Tapped threads shall have a minimum thread engagement in accordance with table IV.

TABLE IV. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.2.6 Welders and welding. Welding and post heat treatment procedures shall be in accordance with the American Welding Society (AWS) welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.2.7 Special tools. The design of the item shall minimize the requirement for special tools (see 7.5.2). All special tools shall be provided with, and stored on, the tow tractor.

3.2.8 Cab specifications.

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3.2.8.1 Cab design. The tow tractor shall be equipped with a front-mounted, enclosed, two-man cab. Doors shall be constructed of rigid materials. The first entrance step to the cab shall not exceed 18.00 inches in height from the ground. If there is more than one step to the cab (including the cab floor), then each step shall be offset out from the next step up by at least 3.00 inches. Steps shall be designed to prevent slipping. Cab instrumentation shall be standard commercial installation to include an hour meter, odometer, tachometer, speedometer, voltmeter, engine oil pressure gauge, engine coolant temperature gauge, transmission temperature gauge, and a fuel level gauge. When an air system is used as the power assist for the braking system (see 3.6.6.2) and for the class 2 tow tractor air system (see 3.2.10), cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light. The tow tractor shall be equipped with an Air Conditioning system. The tow tractor cab shall be equipped with wipers, a windshield washer, driver and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, ventilating fan, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The driver and passenger seats shall incorporate a scissors-type suspension for driver and passenger comfort. The driver seat shall be provided with a safety switch that shuts down the engine in the event the operator leaves the driver seat while the transmission is in forward or reverse gear. The cab shall include a warning light that illuminates as the engine shuts down, and defaults to off once the condition is corrected. The cab shall feature a rear sliding window and a 12V power outlet (cigarette lighter type).

3.2.8.2 Cab visibility. The tow tractor shall be designed to provide maximum visibility in all directions for the range of persons represented between the 5th percentile female driver and the 95th percentile male driver in accordance with table B-IV of MIL-STD-1472. The front and rear couplers shall be visible from the driver position either directly or via a single mirror mounted on the front and rear of the tow tractor.

3.2.8.3 Cab noise levels. Interior cab sound level shall not exceed 84 decibel adjusted (dB(A)) with doors and windows closed. If the sound level is above 84 dB(A), the contractor shall provide a hazardous noise warning plate stating that operator's hearing protection is required when operating this tow tractor. The plate shall be made of a corrosion resistant material and shall be at least 4.00 square inches. It shall have black lettering on a yellow background and shall be located to be visible as the operator enters the cab or is seated. Sound level shall not exceed 94 dB(A).

3.2.9 Tires and wheels. The tow tractor shall be equipped with steel disc wheels and tubeless steel belted radial tires with an appropriate tread pattern. The valve stems may be rubber or metal. Tire valve stem caps shall be plastic. The wheels and tires shall be in accordance with the Tire and Rim Association recommendations for the application. All wheels and tires shall be uniform. If tire loads and tire inflation pressures that exceed the specifications on the tire sidewall markings are required, a copy of the application approval from the tire manufacturer shall be documented in the Technical Manual.

3.2.10 Air system. When specified (see 7.3), the class 2 tow tractor shall have an air system for towing the Air Force's four-wheel MHU-196 and MHU-204 munitions trailers with air brakes. The air system shall be an over-the-road commercial truck/tractor design to include primary and secondary reservoir tanks, a wet tank, heated and automatic drain valves, automotive compressor, dryer, trailer protection valve, low-pressure indicator, service and emergency glad hands, and glad hand covers with lanyards. The glad hands shall be in accordance with SAE J318 and shall be polarized and identified as "service" and "emergency." Compressor sizing shall be

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in accordance with compressor manufacturer's recommendations for off-highway use. Components of the air system that are sensitive to the elements (rainwater, road water, ice build-up, road grit, etc.) and installed in positions on the tow tractor that are vulnerable to those elements shall be covered or shielded to prevent the premature failure of those components over time or during service.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The tow tractor shall be capable of operating in ambient temperatures ranging from –20 to 125 °F. The tow tractor shall be capable of operating in ambient temperatures to –40 °F when heavy-duty (type A) winterization system is specified (see 7.3). Any requirements for towing in high ambient temperatures (recommended coolant ratios, oil types and viscosities, or other fluid types, for various temperature ranges) shall appear in the Technical Manual.

3.3.2 Storage temperature. The tow tractor shall be capable of being stored in ambient temperature to –65 °F when heavy-duty (type A) winterization system is specified (see 7.3).

3.4 Gross Vehicle Weight (GVW) and dimensions. Overall weight and maximum dimensions shall be in accordance with Table V:

TABLE V. GVW and dimensions.

GVW	19,000 pounds
Length	200.00 inches, without couplers
Width	100.00 inches, without mirrors
Height	100.00 inches

3.5 Transportability.

3.5.1 Air transportability. The tow tractor shall be transportable on C-130, C-17 and C-5 aircraft in accordance with MIL-STD-1791. In all air transport configurations, the tow tractor shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The tow tractor shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The tow tractor shall drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.1.1 Shoring. The use of shoring to load the aircraft (referred to as approach shoring and step-up shoring) and during flight (referred to as load shoring, parking shoring, and sleeper shoring) is permitted, but not desired. The use of shoring, especially in loading the aircraft, is not an alternative to a good tow tractor design. The tow tractor shall be designed to the maximum extent possible with adequate clearances for the tow tractor to drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.1.2 Axle weight. Single axle weight shall not exceed 13,000 pounds.

3.5.1.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) for air transport.

3.5.1.4 Air transportable configuration. The air transportable configuration shall include all

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equipment, three-fourth tank of fuel, and no crewmembers. At least 6.00 inches sidewall and 6.00 inches overhead clearance shall be maintained between the tow tractor and the aircraft at all times during loading and flight. The restrained tow tractor shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

3.5.1.5 Equipment removal and reconfiguration. Preparation and restoration of the tow tractor for air transportability shall take no more than 30 minutes for two persons using common non-powered hand tools. All equipment removed shall be stored on the tow tractor. Caps and plugs shall permit driving and storage in transport configuration.

3.5.2 Tie downs. The tow tractor shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the tow tractor. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.5.3 Lifting provisions. The tow tractor shall be equipped with sufficient attachment points so located that it can be lifted by crane. Each attachment point shall be marked "Lift Point." The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

3.6 Performance.

3.6.1 Mobility. The tow tractor without a trailing load shall be capable of traveling a minimum of 25 miles per hour (mi/h) forward and 2.5 mi/h in reverse on flat asphalt or concrete surfaces. The tow tractor shall be capable of towing:

a. MHU-196 munitions trailers weighing 40,000 pounds at 20 mi/h on flat asphalt or concrete surfaces.

b. MHU-196 munitions trailers weighing 80,000 pounds at 10 mi/h on flat asphalt or concrete surfaces.

c. MHU-196 munitions trailers weighing 80,000 pounds at 8 mi/h on 4% grade asphalt or concrete surfaces.

d. C-130 aircraft weighing 175,000 pounds at 5 mi/h on flat concrete surfaces.

3.6.2 Drawbar pull force (DBPF). The tow tractor shall have a minimum DBPF of 16,000 pounds on dry level concrete (coefficient of friction of 0.8).

3.6.3 Suspension. The tow tractor shall have a suspension that will allow for articulation of one or both of the tow tractor axles so that an axle position relative to the other will allow for four-wheel contact with the normal working surfaces found in the above described conditions.

3.6.3.1 Suspension lubrication system. The tow tractor shall be equipped with a lubrication system that automatically and manually provides lubricant to several points on the suspension of the tow tractor, including kingpins, ball joints, steering cylinders, and articulation bolsters. The automatic lubrication system components shall include a pump with integral reservoir, lubricant injectors, tubing, and fittings for the lubrication points.

3.6.4 Vehicle performance.

3.6.4.1 Vehicle maneuvering performance. The class 2 tow tractor, while in the two-wheel mode of steering, shall not have an unsafe tendency to tip, tilt, yaw, sway, skid, or jackknife while towing the MHU-196/204 munitions trailer loaded in the maximum operational configuration at

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maximum operational speeds and while undergoing maximum performance maneuvers such as obstacle avoidance.

3.6.4.2 Air system performance. The class 2 tow tractor air system shall be activated by a brake foot pedal, which modulates air to the braking system and supplies air to the trailer with sufficient speed and volume to actuate the tow tractor and trailer brakes simultaneously. The tow tractor/MHU-196/204 munitions trailer combination shall stop straight in a 12-foot wide corridor. The tow tractor shall not exhibit tendencies to tip, tilt, yaw, sway, skid, bounce, or jackknife under maximum performance maneuvers (emergency braking) in such a way that safe control of the trailer payload is in jeopardy. See 7.6.2 for useful MHU-196/204 munitions trailer specifications.

3.6.5 Driveline and steering. The tow tractor shall have four-wheel drive and two-wheel and four-wheel power steering. Axles shall be planetary-drive/steer axles. Four-wheel steering shall feature both crab and coordinated steering. The selection of the steering mode shall be accomplished by the movement of a one-hand-operated lever or switch. When switching from the crab or coordinated four-wheel mode of steering to the two-wheel mode of steering, the rear wheels shall center automatically (wheels perpendicular to axle) as the steering wheel is rotated to the wheel center, without additional operator input or adjustment. The tow tractor shall have a maximum wall-to-wall turning radius of 17.00 feet or less while in four-wheel coordinated steering mode. Wheel-wells shall be designed so that tire chains may be used without interference.

3.6.6 Brakes.

3.6.6.1 Parking brake. The tow tractor shall be equipped with a disc-type parking brake located on the transmission output shaft. A lockout system shall be provided that disallows the transmission to go into forward or reverse gear, or shuts the engine down, or disallows the engine to respond to throttle when the parking brake is on and the tow tractor gear selector is placed in forward or reverse gear. The lockout system shall also include a cab-mounted warning light. The tow tractor parking brake shall be capable of holding the tow tractor and a fully loaded MHU-141 munitions trailer on an 11.5-degree incline, whether headed up or down the incline, without engagement of the trailer parking brake. The MHU-141 munitions trailer has a maximum loaded weight of 8,400 pounds and a manual parking brake located on the trailer.

3.6.6.2 Service brakes. The tow tractor shall feature a four-wheel, hydraulically operated braking system with power assist. The brakes shall be capable of stopping both the tow tractor without a trailing load from 25 mi/h on flat asphalt or concrete surfaces and the tow tractor with a 175,000 pound towed load on a dry level asphalt or concrete surface from 5 mi/h in no more than 18.00 feet. If an air system is used as the power assist (air-over-hydraulic brakes), cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light (see 3.2.8.1). For the class 2 tow tractor, the air system shall be used as the power assist for the braking system (air-over-hydraulic brakes). Air-over-hydraulic braking systems shall be activated by a brake foot pedal, which modulates air to the braking system.

3.6.7 Couplers. The tow tractor shall be equipped with front and rear clevis-pin type couplers. The couplers shall accommodate lunettes (drawbar rings) with a 3.00 inch inside diameter and 6.25 inch outside diameter. The front and rear couplers shall be located on the centerline of the tow tractor with the coupling positions located at 25.00 inches, plus or minus 0.50 inch, above the ground. Additional mounting holes (front and rear) for couplers with the coupling positions

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located at 14.00 inches, plus or minus 0.50 inch, above the ground shall be provided. Coupling positions are measured to the centerline of lunette (drawbar ring) as it rests within the coupler. The lunette has a material diameter or height of 1.63 inches.

a. The preferred coupler is manufactured by SAF-Holland as a Holland EH-3050 E-Hitch. The couplers shall be attached to the tow tractor using a quantity of (2), ½-13, grade 8 fasteners tightened to a torque value of 108 lb·ft (dry) in either the upper set or the inner set of 0.50-inch hole positions, and a quantity of (2), ⅝-11, grade 8 fasteners tightened to a torque value of 212 lb·ft (dry) in the lower set of 0.625-inch hole positions found on the Holland EH-3050 E-Hitch design. Fastener tightening torque is per industry-accepted torque tables that preloads the fastener (producing a clamp load) that is 75% of the proof load of the fastener to prevent losing or fatiguing the fasteners and/or the coupler, while maintaining an adequate safety factor for the fastener. A description of the coupler fastener type and the fastening torque shall be documented in the Technical Manual. The EH-3050 E-Hitch has been previously certified to meet the coupler strength requirements.

b. If the preferred coupler is not available, the tow tractor manufacturer shall manufacture a coupler to the following design specifications:

    1) The clevis shall be constructed from ASTM-A36/A36M steel material.

    2) The clevis may be cast or welded construction. If the clevis is welded construction, one edge on each of the upper and lower plates, and two edges of the mounting plate shall be beveled at 0.375 inch, such that the weld joint locations at the top and bottom surfaces of the assembled clevis are beveled, prior to welding. All joints shall be fully welded with 0.375-inch welds. The welded clevis assembly shall be heat treated for stress relief per AWS D1.1/D1.1M:2015, paragraphs 5.8 and 5.8.1.

    3) The upper and lower plates of the clevis shall have the shape of an isosceles triangle with a 9.00-inch base. The centerline of a 2.25-inch diameter thru-hole shall be located at 9.00 inches from the center point of the base. A 3.00-inch radius extending from the centerline of the thru-hole shall define a circular edge of the plates, which shall merge tangentially with a line from each end point of the base. The upper and lower plates of the clevis shall be 1.00 inch thick and separated by 3.00 inches of clearance.

    4) The mounting plate of the clevis shall be 1.00 inch thick by 5.00 inches high by 10.00 inches wide. The bolt pattern for the fasteners on the mounting plate shall consist of a quantity of (4) 0.688-inch diameter thru-holes, arranged on a rectangular pattern, spaced at 1.75 inches high by 3.375 inches wide, centerline of hole to centerline of hole, and situated symmetrically from top to bottom left to right on the mounting plate. Fasteners shall be accessible with a socket and extension to allow for tightening with a standard torque wrench.

    5) A vertical gusset measuring 0.75 inch thick by 3.00 inches high by 5.00 inches deep shall separate the upper and lower plates along the hitch centerline and shall also attach to the mounting plate.

    6) The couplers shall be attached to the tow tractor using a quantity of (4), ⅝-11, grade 8 fasteners tightened to a torque value of 212 lb·ft (dry). Fastener tightening torque is per industry-accepted torque tables that preloads the fastener (producing a clamp

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load) that is 75% of the proof load of the fastener to prevent losing or fatiguing the fasteners and/or the coupler, while maintaining an adequate safety factor for the fastener. A description of the coupler fastener type and the fastening torque shall be documented in the Technical Manual.

7) The pin shall be constructed from 8620 steel. The pin shall be 2.25 inches in diameter with a 30-degree taper along 0.25 inch on the end, case-hardened to Rc 55-60 at a depth of 0.060 inch for wear resistance and have a pull handle to lift and place the pin into the clevis. The pin shall be coated with a zinc finish. Pin length shall be optimized so as not to interfere with terminal connection of the J560 electrical receptacle (see 3.8.3).

8) The coupler shall incorporate a spring-loaded pin retention device to prevent the pin from bouncing out of the hitch while towing. The spring shall be stainless steel.

9) The coupler shall incorporate an offset-mounted ring or tube on the left side as a holding device for the pin, and a chain lanyard to prevent the loss of the pin. The length and attachment point of the chain lanyard shall be optimized to prevent the chain from hanging into the coupled area when towing.

10) The coupler shall allow for a minimum turning angle of 90 degrees on each side of the coupler centerline relative to the drawbar centerline.

c. The tow tractor front and rear coupler and mounting structure shall be designed to withstand the following load forces:

1) A tension force value of 16,000 pounds applied in the same vector direction as DBPF in a straight pull configuration and a pull at 45 degrees on each side of the centerline.

2) A compression force value of 16,000 pounds applied in the same vector direction as DBPF in a straight push configuration and a push at 45 degrees on each side of the centerline.

3) These loads shall be multiplied by a safety factor of 2.0 and shall be the minimum loads for attaining the design stress levels.

3.7 Engine and related equipment.

3.7.1 Engine. The tow tractor shall have a commercially available liquid-cooled, turbocharged diesel engine of not less than six cylinders and power train sufficient to develop the required drawbar pull and obtain the required tow tractor speeds. The diesel engine shall operate on high-sulfur diesel fuels (containing over 15 parts per million (ppm) sulfur) and ultra-low sulfur diesel fuels (containing less than 15-ppm sulfur). At a minimum, the tow tractor shall meet a Tier III level exhaust and evaporative emissions standard (see 7.6.1) for industrial (off-highway) engines. The engine shall include an anti-restart and a neutral safety start feature. Engine shall have an automatic engine shutdown system for high coolant temperature, low oil pressure, and engine over speed (engine in excess of 110% of rated speed).

3.7.2 Engine starting aids. The engine shall start, within 15 seconds, cranking in any ambient temperature within the required operating range of the tow tractor. Internal engine starting aids, fluid starting aids, and heat from the winterization system (see 3.7.6) may be used prior to and during the start period to facilitate engine starting under the following conditions in table VI:

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TABLE VI. Engine starting aids.

Temperature Range	Starting Aids Permitted
40 through 125 °F	None
0 through 39 °F	Internal engine starting aids and fluid starting aids
–40 through –1 °F	Internal engine starting aids, fluid starting aids, and heat from the winterization system

3.7.3 Engine cooling system. The tow tractor engine shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of 125 °F without exceeding the manufacturer's maximum allowable temperatures on the individual engine components.

3.7.4 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. The exhaust system outlet shall exit on the right-hand side over the rear axle and behind the rear wheel, oriented at 90 degrees from the centerline of the tow tractor and not directed toward the ground.

3.7.5 Transmission. Transmission shall be of the power-shift type with an integral limited-slip, inter-axle differential. Maximum allowable transmission temperature during all operations shall not exceed 260 °F.

3.7.6 Winterization system. A standard (type C) winterization system shall be provided for operation in temperatures to –20 °F. When specified (see 7.3), a heavy-duty (type A) winterization system shall be provided for operation to –40 °F and storage to –65 °F. The winterization systems may include heaters for engine coolant, engine oil, fuel, and battery warmers. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. For each winterization system, a breakdown of the winterization components and requirements (recommended coolant ratios, oil types and viscosities, or other fluid types for the various temperature ranges) shall be documented in the Technical Manual.

3.7.6.1 Winterization plugs and connectors. Winterization package components with external power requirements shall be compatible with 110-volt (V) alternating current (AC), 60-hertz (Hz) power source utilizing covered male inlet plugs in accordance with National Electrical Manufacturers Association (NEMA) WD 6.

3.8 Electrical system. The tow tractor shall be equipped with a 12V, negative ground electrical system. All external chassis wiring shall incorporate weatherproof connections and terminations. Exceptions to this are under-hood terminal strips and electrical posts that are protected from the elements. Any wire splicing (typically resulting from special-request options) shall incorporate a mechanically crimped connector, solder, and heat-shrink tubing. Crimped-on wire connectors shall be installed with a corresponding crimping tool. All wiring shall be sized to handle the system voltage and current levels. Battery cables shall be identified with "+" and "-" or red and black markings and shall not be spliced.

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3.8.1 Lighting. The tow tractor lighting shall be equivalent to or exceed the standard commercial offering to include a rear adjustable aim floodlight, an adjustable spotlight, parking lights, head lights (high and low beam), brake lights, turn signals, hazard lights wired to a constant-hot circuit, and reverse lights. A flashing amber beacon light shall be mounted on the cab roof on the tow tractor centerline. The specified lights shall be constructed with Light-Emitting Diodes (LED). The rear adjustable aim floodlight of not less than 3000 candlepower shall be mounted on the tow tractor so that the following areas are illuminated: the rear working area behind the cab, the rear coupling positions, and a reflective, vertical indicator stripe measuring approximately 1.00 inch x 8.00 inches located on the towed MHU-196/204 munitions trailer. The stripe is located at a horizontal distance of 10.00 feet behind the coupler and a vertical distance of 2.00 feet 2.00 inches to 2.00 feet 10.00 inches from the ground. The adjustable spotlight, capable of illuminating 180 degrees of elevation and 360 degrees of rotation, shall be centrally located on the top, rear portion of the cab. The spotlight shall be installed in the cab's roof and shall be adjustable from the driver position, inside the cab. The primary function of the spotlight is to illuminate the towing path of an aircraft, rear of the tow tractor, but the spotlight shall also be capable of illuminating areas in the front and sides of the tow tractor.

3.8.2 Batteries and battery compartment.

3.8.2.1 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

3.8.2.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

3.8.3 12-volt trailer light harness receptacle. The rear of the class 2 tow tractor shall be equipped with an SAE J560 electrical receptacle, spaced equally between the upper and lower coupler positions but offset 6.00 inches to the left of the tow tractor centerline. All terminals of the receptacle shall be wired in accordance with SAE J560. The #7 terminal shall be supplied with an ignition-hot battery source, with a minimum circuit protection of 10 amps.

3.9 Towing safety attachments.

3.9.1 Breakaway cable hookup. For the class 2 tow tractor, the rear of the tow tractor shall be equipped with an emergency trailer-breakaway cable attachment point, capable of mating with a National Aerospace Standards (NAS) 1281C10 spring snap hook. The attachment point shall be located on the right hand side and within 5.00 inches of the SAE J560 electrical receptacle (see 3.8.3). The attachment point shall not be installed on the coupler.

3.9.2 Safety chain attachment points. For the class 2 tow tractor, the rear of the tow tractor shall be equipped with two safety chain attachment points for use when towing, composed of hinged D-rings suspended in the downward position. The D-rings shall be compatible with trailer safety chain hooks with an inside radius between 1.00 and 3.00 inches. The D-rings shall be located symmetrically on each side of the lower coupler position at a distance of 14.00 inches above the ground. The vertical centerline of the D-rings shall be located at a parallel distance of no greater than 12.00 inches to the left and right of the tow tractor's centerline. The D-rings shall not be installed on the coupler assembly. Each D-ring shall be capable of withstanding a shock load of 16,000 pounds and shall have the ability for standard safety chain hooks to be attached.

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3.10 Diagnostic software. When specified (see 7.3), a copy of any diagnostic software on CD-ROM or DVD-ROM and/or an accessory module required or recommended for maintaining the tow tractor shall be provided.

3.11 Workmanship. The tow tractor, including all parts and accessories, shall be constructed and finished in a thoroughly competent manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.11.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.11.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.11.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.11.4 Cleaning. The tow tractor shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states that the GPP requires 100% of EPA-designated product purchases (that are included in the Comprehensive Procurement Guidelines list) contain recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe, b) meeting appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the

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commercial marketplace, modified as necessary to comply with the requirements herein. The government reserves the right to require proof of such conformance.

5.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Pre-production inspection (see 5.3).

b. Conformance inspection (see 5.4).

5.3 Pre-production inspection. The pre-production tow tractor shall be subjected to demonstrations, examinations, and tests and supplied with analyses, certifications, and lists of components as described in 5.6.1 through 5.6.31.3. The contractor shall provide or arrange for all test equipment and facilities.

5.4 Conformance inspection. Each production tow tractor shall be subjected to the examination described in 5.6.1.

5.5 Inspection requirements.

5.5.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.5.2 Data. During all testing specified herein, at least the following data shall be recorded. Additional data shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

5.5.3 Test rejection criteria. Throughout all tests specified herein, the tow tractor shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the Original Equipment Manufacturer (OEM) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the tow tractor components or between the tow tractor, the ground, and all required obstacles, with the exception of normal contact by the tires.

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i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

1) Engine cooling system.

2) Engine lubrication system.

3) Engine protective circuits.

5.6 Test methods.

5.6.1 Examination of product. Each tow tractor shall be examined to verify compliance with the requirements. The pre-production tow tractor (see 5.3) shall be examined using a contractor-generated, government-approved test procedure to identify each requirement verified by demonstration, examination, or test. Attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each tow tractor function shall be verified. Analyses, certifications, lists of components, and schematics shall be provided in accordance with table VII. Each production tow tractor (see 5.4) shall be examined using a contractor-generated, government-approved checklist that is a reduced version of the pre-production checklist.

TABLE VII. Analyses, certifications, lists of components, and schematics.

Paragraph	Required Analyses, Certifications, Lists of Components, and Schematics
3.2 Design and construction.	Contractor certification (see 5.6.2)
3.2.10 Air system.	Contractor schematic (see 5.6.11)
3.3.1 Operating temperature range.	Contractor certification (see 5.6.12.1)
3.3.2 Storage temperature.	Contractor certification (see 5.6.12.2)
3.5.1 Air transportability.	Contractor air transportability analysis (see 5.6.14.1)
3.5.1.1 Shoring.	
3.5.1.4 Air transportable configuration.	
3.5.2 Tie downs.	Contractor tie down provision analysis (see 5.6.14.3)
3.5.3 Lifting provisions.	Contractor lifting provision analysis (see 5.6.14.4)
3.6.5 Driveline and steering (axles).	Axle manufacturer certification (see 5.6.19.1)

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TABLE VII. Analyses, certifications, lists of components, and schematics – Continued.

Paragraph	Required Analyses, Certifications, Lists of Components, and Schematics
3.6.6.2 Service Brakes.	Contractor schematic (see 5.6.20.2)
3.6.7 Couplers.	Contractor analysis (see 5.6.21).
3.7.1 Engine.	Engine manufacturer certification (see 5.6.22)
3.7.3 Engine cooling system.	
3.7.5 Transmission.	Transmission manufacturer certification (see 5.6.22)
3.7.6 Winterization system.	Contractor list of components and system requirements (see 5.6.24)
3.8.3 12-volt trailer light harness receptacle.	Contractor electrical schematic (see 5.6.28)
3.10 Diagnostic software.	Contractor list of components (see 5.6.30)

5.6.2 Design and construction certification. A contractor certification shall be submitted to the government representative at the pre-production inspection stating that the tow tractor has been designed and constructed per SAE specifications ARP1247 and AIR1363, in compliance with the requirements of 3.2.

5.6.3 Finish examination. The pre-production tow tractor shall be examined for verification of the anti-skid material on the top surfaces of the tow tractor, in compliance with the requirements of 3.2.1.2.

5.6.4 Corrosion protection examination. The pre-production tow tractor shall be examined for verification of the permanent CPC, in compliance with the requirements of 3.2.1.4.

5.6.5 Markings examination. The pre-production tow tractor shall be examined for verification of markings, in compliance with the requirements of 3.2.2.

5.6.6 Identification and data plates examination. The pre-production tow tractor shall be examined for verification of the identification and data plates, in compliance with the requirements of 3.2.3.1, 3.2.3.2, and 3.2.3.3.

5.6.7 Fire extinguisher examination. The pre-production tow tractor shall be examined for verification of the fire extinguisher, in compliance with the requirements of 3.2.4.3.

5.6.8 Back-up alarm examination. The pre-production tow tractor shall be examined for verification of the back-up alarm, in compliance with the requirements of 3.2.4.4.

5.6.9 Cab verification.

5.6.9.1 Cab design examination. The pre-production tow tractor shall be examined for verification of cab design, in compliance with the requirements of 3.2.8.1.

5.6.9.2 Cab visibility examination. The pre-production tow tractor shall be examined for verification of cab visibility, in compliance with the requirements of 3.2.8.2.

5.6.9.3 Cab noise level test. The pre-production tow tractor shall be tested for verification of compliance with the requirements of 3.2.8.3 by measuring the noise level at the operator's ear with the engine operating at maximum rpm and windows and doors closed. If noise level is

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between 84 and 94 dB(A), the noise warning plate required by 3.2.8.3 shall be on all tow tractors delivered on this contract.

5.6.10 Tires and wheels examination. The pre-production tow tractor shall be examined for verification of tires and wheels, in compliance with the requirements of 3.2.9.

5.6.11 Air system schematic. A schematic of the air system for the class 2 tow tractor shall be submitted to the government representative at the pre-production verification, for verification of compliance with the requirements of 3.2.10.

5.6.12 Environmental conditions verification.

5.6.12.1 Operating temperature range certification. A contractor certification shall be submitted to the government representative at the pre-production inspection stating that the tow tractor and all temperature sensitive components are capable of operation in the specified temperature ranges, in compliance with the requirements of 3.3.1.

5.6.12.2. Storage temperature certification. A contractor certification shall be submitted to the Government representative at the pre-production inspection stating that the tow tractor and all temperature sensitive components are capable of storage to the specified temperature, in compliance with the requirements of 3.3.2.

5.6.13 Weight and dimension tests.

5.6.13.1 Weight test. The pre-production tow tractor shall be weighed for verification of the GVW and axle weights, in compliance with the GVW requirement of 3.4 and the single axle weight requirement of 3.5.1.2.

5.6.13.2 Dimension measurement test. The pre-production tow tractor shall be measured for verification of the dimensions, in compliance with the requirements of 3.4.

5.6.14 Transportability verification.

5.6.14.1 Air transportability analysis. An engineering analysis shall be performed for verification of compliance with the air transportability requirements of 3.5.1, 3.5.1.1, and 3.5.1.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.5.1. The engineering analysis shall also include a dimensional analysis for the tow tractor while traversing the ramp and while loaded aboard C-17 and C-5 aircraft.

5.6.14.2 Equipment removal and reconfiguration demonstration. For verification of compliance with the requirements of 3.5.1.5, the pre-production tow tractor shall be configured for transport on C-17 and C-5 aircraft and then reconfigured for operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

5.6.14.3 Tie down provision analysis. An engineering analysis shall be performed and submitted to the government representative at the pre-production inspection for verification of compliance with the tie down provision requirements of 3.5.2.

5.6.14.4 Lifting provision analysis. An engineering analysis shall be performed and submitted to the government representative at the pre-production inspection for verification of compliance with the lifting provision requirements of 3.5.3.

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5.6.15 Mobility test. The pre-production tow tractor shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the tow tractor speedometer as the speed-measuring device, in compliance with the requirements of 3.6.1.

5.6.16 Drawbar pull test. The pre-production tow tractor shall be tested for verification of the DBPF on dry level concrete using a load-sensing device and a fixed anchoring position in compliance with the requirements of 3.6.2. The drawbar position shall be horizontal and level.

5.6.17 Suspension examination. The pre-production tow tractor shall be examined for verification of an articulating suspension in compliance with the requirements of 3.6.3.

5.6.18 Suspension lubrication system examination. The pre-production tow tractor shall be examined for verification of a lubrication system for the suspension, in compliance with the requirements of 3.6.3.1.

5.6.19 Driveline and steering verification.

5.6.19.1 Axles certification. For verification of compliance with the requirements of 3.6.5, a manufacturer certification shall be submitted to the government representative at the pre-production inspection stating that the axles are the correct application for the torque and power ratings of the tow tractor at the specified temperature ranges.

5.6.19.2 Driveline and steering test. The pre-production tow tractor shall be tested in two-wheel steer and four-wheel steer, using both crab and coordinated steering for verification of compliance with the requirements of 3.6.5. The tow tractor shall be tested for verification of tire chain use without interference in wheel-wells. Tire chains shall be fitted to the tires, and the tow tractor shall be operated by driving a short distance and turning the wheels to the extreme left and right. The tow tractor shall be tested for verification of the maximum wall-to-wall turning radius while in four-wheel coordinated steering mode. The fully ballasted pre-production tow tractor shall be tested as follows for verification of compliance with the turning radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the tow tractor being tested.

b. Equipment required.

    1) A device suitable for measuring three times the length of the tow tractor being tested with an accuracy of at least ±1.00 inch.

    2) Markers or marking device suitable for marking the pavement.

    3) A plumb bob suitable for locating a point on the pavement directly below a fixed point on the tow tractor.

    4) A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.

c. Test Conditions.

    1) The tow tractor shall be fully ballasted and in operational form.

    2) The tow tractor steering system shall be fully operational and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.

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d. Test Procedure.

1) The tow tractor shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.

2) Continue driving the slow full cramp circle.

3) At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the tow tractor using the service brakes.

4) t each stop, place a plumb bob against the outermost point of the tow tractor and mark the spot on the ground directly below where the plumb bob comes to rest.

5) Measure and record the straight line distances between each pair of points, that is, Lengths AB, BC, and CA.

6) Calculate the wall-to-wall turning radius (R) as follows:

$$R = D / 2 \quad \text{where}$$

$$D = \frac{(AAAAxAABBxBBAA)}{2x\sqrt{[SSx(SS-AAAA)x(SS-AABB)x(SS-BBAA)]}} \text{ and}$$

$$S = (AB+BC+CA) / 2$$

e. Repeat steps (i) through (vi) with the tow tractor moving in the opposite direction.

5.6.20 Brakes verification.

5.6.20.1 Parking brake examination and test. The pre-production tow tractor shall be examined for verification of the parking brake, examined and tested for the parking brake lockout system, and tested using the tow tractor parking brake only with a towed 8,400-pound load on an 11.5-degree incline, with tow tractor positioned both up and down the incline to demonstrate compliance with the requirements of 3.6.6.1.

5.6.20.2 Service brake schematic, examination, and test. For verification of compliance with the requirements of 3.6.6.2, a schematic of the braking system for the class 1 and 2 tow tractors shall be submitted to the government representative at the pre-production verification. The pre-production tow tractor shall be examined for verification of a four-wheel, hydraulically operated braking system with power assist and tested for verification of the service brake operation from 25 mi/h on a flat asphalt or concrete surface without a trailing load, in compliance with the requirements of 3.6.1 and 3.6.6.2.

5.6.21 Coupler and mounting structure analysis and examination. An engineering analysis report shall be performed and submitted to the government representative at the pre-production inspection for verification of compliance with the strength requirements of 3.6.7. The report shall contain supporting calculations to show that the front and rear coupler and mounting structures are in compliance with their respective strength specifications. The rear coupler and mounting structure calculations shall be supplemented with a Finite Element Analysis (FEA) to further support compliance with the strength specifications. A finite-element analysis of the tow tractor rear coupler and mounting structure shall be submitted to the government representative at the pre-production inspection, for verification of compliance with the requirements of 3.6.7. The pre-production tow tractor shall be examined for verification of the couplers, in compliance with the requirements of 3.6.7.

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5.6.22 Engine and related equipment certification. A manufacturer certification shall be submitted to the government representative at the pre-production inspection stating that the engine and transmission is in compliance with the requirements of 3.7.1, 3.7.3, and 3.7.5, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including cooling system, lubrication system, and mounting system, shall be included in the certification.

5.6.23 Exhaust system examination. The pre-production tow tractor shall be examined for verification of the exhaust system, in compliance with the requirements of 3.7.4.

5.6.24 Winterization system, list of components and system requirements, and examination. A list of components for the winterization system and requirements (recommended coolant ratios, oil types and viscosities, or other fluid types for the various temperature ranges) for each winterization system shall be submitted to the government representative at the pre-production inspection. Reference the requirements of 3.7.6. The pre-production tow tractor shall be examined for a winterization system (either type C or type A) in compliance with the requirements of 3.7.6 and 3.7.6.1.

5.6.25 Electrical system examination. The pre-production tow tractor shall be examined for verification of an electrical system, in compliance with the requirements of 3.8.

5.6.26 Lighting examination. The pre-production tow tractor shall be examined for verification of lighting, in compliance with the requirements of 3.8.1.

5.6.27 Batteries and battery compartment examination. The pre-production tow tractor shall be examined for verification of batteries and battery compartment, in compliance with the requirements of 3.8.2.1 and 3.8.2.2.

5.6.28 12-volt trailer light harness receptacle schematic, examination, and test. An electrical schematic of the 12-volt trailer light harness circuitry and its interface with the OEM wiring shall be submitted to the government representative at the pre-production inspection, for verification of compliance with the requirements of 3.8.3. The pre-production tow tractor shall be examined for verification of the 12-volt trailer light harness receptacle, in compliance with the requirements of 3.8.3. Each circuit in the receptacle shall be tested with a light tester or a voltmeter to confirm circuit function when actuated, in compliance with the requirements of 3.8.3.

5.6.29 Towing safety attachments.

5.6.29.1 Breakaway cable hookup examination. The pre-production tow tractor shall be examined for verification of the breakaway cable hookup and location, in compliance with the requirements of 3.9.1.

5.6.29.2 Safety chain attachment points examination. The pre-production tow tractor shall be examined for verification of the safety chain attachment points and location, in compliance with the requirements of 3.9.2.

5.6.30 Diagnostic software, list of components. A list of components for the diagnostic software to be supplied shall be submitted to the government representative at the pre-production inspection, in compliance with the requirements of 3.10.

5.6.31 Compatibility tests.

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5.6.31.1 Mobility tests with towed loads. The pre-production class 2 tow tractor shall be tested for verification of its capability to tow configurations (a), (b), and (c) of the following, using the tow tractor's speedometer as the speed measuring device in compliance with the requirements of 3.6.1 (a), (b), and (c). The pre-production class 1 or class 2 tow tractor shall be tested for verification of its capability to tow configuration d. of the following, using the tow tractor speedometer as the speed measuring device in compliance with the requirements of 3.6.1 d.

a. MHU-196 munitions trailers weighing 40,000 pounds at 20 mi/h on flat asphalt or concrete surfaces.

b. MHU-196 munitions trailers weighing 80,000 pounds at 10 mi/h on flat asphalt or concrete surfaces.

c. MHU-196 munitions trailers weighing 80,000 pounds at 8 mi/h on 4% grade asphalt or concrete surfaces.

d. C-130 aircraft weighing 175,000 pounds at 5 mi/h on flat concrete surfaces. This test may be conducted while towing an alternative towed load weighing 175,000 pounds.

5.6.31.2 Vehicle performance tests. The pre-production class 2 tow tractor shall be tested while towing an Air Force MHU-196 munitions trailer to demonstrate compliance with the vehicle maneuvering performance requirements of 3.6.4.1 and air system performance requirements of 3.6.4.2. Two Air Force tow tractor operators shall perform each of the tests. A follow vehicle shall drive behind the tow tractor/trailer train at a safe distance to monitor the tow tractor performance. Both tests shall be performed in the below three test configurations:

a. Tow tractor without a trailing load.

b. Tow tractor towing a 40,000-pound, unloaded MHU-196 munitions trailer.

c. Tow tractor towing an 80,000-pound, loaded MHU-196 munitions trailer.

5.6.31.2.1 Collision avoidance test. For each test configuration (a), (b), and (c), the pre-production class 2 tow tractor shall be brought up to the specified test speed and subjected to rapid left and right lane changes that simulate the avoidance of collisions. While performing the lane change maneuvers, the equipment shall remain within a 22-feet wide boundary. Test configurations (a) and (b) shall be conducted at 20 mi/h. Test configuration (c) shall be conducted at 10 mi/h. Operators shall accelerate and steer the tow tractor (and trailer) in a cautious and conservative manner in order to gain familiarization with the characteristics of the tow tractor (and trailer), progressing toward increasing speed and steering input/response. Operators shall base the rate of progression of the testing on his/her sense of comfort and control. Eventually, rapid left and right turn lane changes from 20 mi/h (tow tractor without a trailing load and tow tractor towing a 40,000-pound, unloaded MHU-196 munitions trailer) and from 10 mi/h (tow tractor towing an 80,000-pound, loaded MHU-196 munitions trailer) shall be conducted regularly. Once this level of vehicle performance is achieved, each of the two operators shall repeat each test configuration 12 times.

a. Tow tractor without a trailing load.

b. Tow tractor towing a 40,000-pound, unloaded MHU-196 munitions trailer.

c. Tow tractor towing an 80,000-pound, loaded MHU-196 munitions trailer.

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5.6.31.2.2 Air system brake performance test. For each test configuration (a), (b), and (c), the pre-production class 2 tow tractor shall be brought up to the specified test speed and subjected to a maximum braking effort and brought to a complete stop. Operators shall accelerate and brake the tow tractor (and trailer) in a cautious and conservative manner in order to gain familiarization with the characteristics of the tow tractor (and trailer), progressing toward increasing speed and braking input/response. Operators shall base the rate of progression by increasing speed in increments of 5 mi/h. Eventually, maximum braking efforts from 20 mi/h (tow tractor without a trailing load and tow tractor towing a 40,000 pound, unloaded MHU-196 munitions trailer) and from 10 mi/h (tow tractor towing an 80,000-pound, loaded MHU-196 munitions trailer) shall be conducted regularly. Once this level of vehicle performance is achieved, each of the two operators shall repeat each test configuration 12 times. At the conclusion of the tests for configurations (b) and (c), the operators and observers shall be interviewed to gather and document opinions of the tow tractor braking performance. Discussion points shall include brake modulation, smooth vs. rough stopping, and brake "feel" (sufficient air volume and pressure).

a. Test configuration (a) shall be conducted at 20 mi/h. The pre-production class 2 tow tractor shall be brought to the specified speed, subjected to a maximum braking effort, and brought to a complete stop within a distance of 35 feet and within a 12-feet wide lane. The stopping distance of the tow tractor with no trailing load shall be tested and recorded 12 times by each of the two operators.

b. Test configuration (b) shall be conducted at 20 mi/h. The pre-production class 2 tow tractor and unloaded MHU-196 munitions trailer shall be brought to the specified speed and subjected to a maximum braking effort and brought to a complete stop, simulating a panic braking event. Passing test results include minimal amounts of wheel lock-up, skids, bouncing, and yawing (sideslip) from the tow tractor during panic braking, with the tow tractor/trailer train coming to a stop within a 12-feet wide lane. Severe jack knifing shall be classified as failing testing results. Each of the two operators shall repeat test configuration (b) 12 times.

c. Test configuration (c) shall be conducted at 10 mi/h. The pre-production class 2 tow tractor and loaded MHU-196 munitions trailer shall be brought to the specified speed, subjected to a maximum braking effort, and brought to a complete stop, simulating a panic braking event. Passing test results include minimal amounts of wheel lock-up, skids, bouncing, and yawing (sideslip) from the tow tractor during panic braking, with the tow tractor/trailer train coming to a stop within a 12 feet-wide lane. Severe jack knifing shall be classified as failing testing results. Each of the two operators shall repeat test configuration (c) 12 times.

5.6.31.3 Service brake test. The pre-production class 1 or 2 tow tractor shall be tested by an Air Force tow tractor operator for verification of its capability to stop a 175,000-pound C-130 aircraft on a dry level asphalt or concrete surface from 5 mi/h in no more than 18.00 feet to demonstrate compliance with 3.6.6.2. This test may be conducted while towing an alternative towed load weighing 175,000 pounds.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6.1 Technical data.

6.1.1 Manufacturer publications. The contractor shall provide the following manufacturer's publications in Indexed Portable Document Format (IPDF). All data shall be supplied in

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accordance with DFARS clause 252.227-7015. All manuals shall be delivered with either unlimited data rights or as a minimum, a Government Purpose License, to allow the Air Force to electronically store, print, distribute and update the manual. Manuals are subject to government review and approval in accordance with MIL-PRF-32216. Any changes or corrections noted by the Air Force shall be corrected and updated pages or manuals shall be submitted. Only one digital copy is required per order regardless of quantity of units.

6.1.2 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the vehicle. The operator's manual shall include at least the following:

a. General description and step-by-step instructions for the operation of the vehicle and its auxiliary equipment

b. Checklists, lubrication charts, and basic troubleshooting procedures for the daily maintenance inspection and mission readiness checks that the operator is expected to perform

c. Schedules for required preventative maintenance and required periodic maintenance

d. Full descriptions and illustrations for the location and function of all controls

e. List of all checks and adjustments needed in preparation for placing the vehicle into service upon receipt from the contractor

f. Preparation Instructions for shipment or storage

g. Warranty information and a list that details the period of the warranty for the complete vehicle and for any component that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers listed.

6.1.3 Service and repair manuals. Service and repair manuals shall be provided for the chassis, engine, transmission, and equipment. The service manuals shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, fluid capacities, current data, voltage data, resistance data, test procedures, and any illustrations or exploded views that may be required to perform proper maintenance by qualified mechanics. The service and repair manuals shall contain an alphabetical subject index as well as a table of contents. The manual shall contain at least the following where applicable:

a. Pneumatic, electrical, and fuel schematics

b. Schedules for required preventative maintenance and required periodic maintenance

c. Troubleshooting, servicing, disassembly, repair, and assembly procedures

6.1.4 Illustrated parts breakdown. An illustrated parts breakdown (IPB) manual shall be provided for the chassis, engine, transmission, and equipment. IPB manuals shall include illustrations and exploded views as needed to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts list. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The IPB manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The IPB manuals shall contain a numerical index. The manuals shall contain a list of all the

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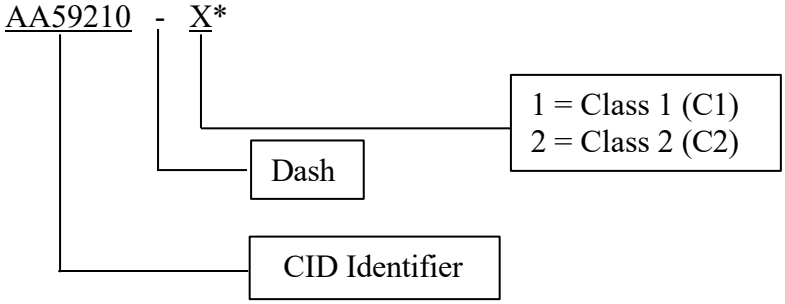
component vendor names, addresses, and telephone numbers referenced in the parts list.

6.1.5 Manual delivery. Digital copies shall be provided via digital file transfer such as Department of Defense Secure Access File Exchange (DOD SAFE) and shall be furnished to AFLCMC/ROZ 30 days prior to the first hardware delivery. Contact the appropriate Integrated Product Team Equipment Specialist for file delivery instructions.

6.1.6 Service bulletins. The contractor shall prepare service bulletins as required to notify the government of new maintenance problems and/or procedures to be implemented on the contractor's equipment prior to the issuance of permanent changes to the technical manuals. Service bulletins shall also be prepared when the program office has identified a trend of a potential problem or failure that has occurred at field level. The service bulletins shall identify any potential safety hazards and shall be submitted in a Microsoft Office digital format.

7. NOTES.

7.1 Part or Identification Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



*This example (AA59210-C1) describes a PIN for CID A-A-59210.

7.2 Source of documents.

7.2.1 Military specifications, standards, and handbooks. referenced herein may be obtained at https://quicksearch.dla.mil.

7.2.2 Federal Acquisition Regulations (FAR). FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/browse/index/far.

7.2.3 American Society for Testing and Materials (ASTM). Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from https://www.astm.org/.

7.2.4 American Welding Society (AWS). Application for copies should be addressed to American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126. Electronic copies of AWS standards may be obtained from https://www.aws.org.

7.2.5 Society of Automotive Engineers (SAE). Application for copies should be addressed to SAE Inc., 400 Commonwealth Drive, Warrendale, PA 15096. Electronic copies of SAE standards may be obtained from https://www.sae.org/.

7.2.6 National Aerospace Standards (NAS). Application for copies should be addressed to Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209. Electronic copies of NAS standards may be obtained from https://www.aia-aerospace.org/.

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7.2.7 National Electrical Manufacturers Association (NEMA). Application for copies should be addressed to National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1752, Rosslyn, Virginia 22209. Electronic copies of NEMA standards may be obtained from https://www.nema.org/stds/.

7.2.8 Cortec documents. Application for copies should be addressed to 4119 White Bear Parkway St. Paul, MN 55110 U.S.A. Electronic copies of Cortec standards may be obtained from https://www.cortecvci.com.

7.2.9 Tire and Rim Association (TRA) Standards. The Tire and Rim Association, Inc. (TRA) standards may be obtained from https://www.us-tra.org/. or from the Tire and Rim Association, 175 Montrose West Ave., Suite 150, Copley, OH 44321.

7.2.10 Code of Federal Regulations eCFR. Copies of eCFR documents may be obtained at https://www.ecfr.gov/.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Packaging requirements (see 6.).

c. Finish color required (gloss white) (see 3.2.1.2).

d. Temporary CPC (see 3.2.1.4).

e. Winterization A (see 3.3.1, 3.3.2, and 3.7.6).

f. Class 2 with air system for trailer towing (see 3.2.10).

g. Diagnostic software (see 3.10).

7.4 National Stock Number (NSN). The following Table VIII is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

TABLE VIII. NSN.

NSN	Class	PIN
1740-00-580-7990	1	AA59210-C1
1740-01-617-5631	2	AA59210-C2

7.5 Definitions.

7.5.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1.00 inch and metric sizes up to and including 25 mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

7.5.2 Special tool. A tool that is not commercially and readily available from a source other than the tow tractor contractor.

7.6 Additional information.

7.6.1 Engine waiver. Tier III "off-highway" represents the highest level of exhaust and evaporative emissions standards that will permit the use of high sulfur fuels. The Air Force possesses an Environmental Protection Agency (EPA) waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a tag or decal for

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the engine stating “insert engine description here – This engine has an exemption for national security under 40 CFR 89.908 and/or 40 CFR 1068.225.”

7.6.2 MHU-196/204 specifications. The MHU-196/204 munitions trailers have one-fourth inch inner diameter (ID) service brake air lines and three-eighth inch ID parking brake air lines. The trailers have air brake chambers and drum brakes on each wheel, and a complete self-contained air control system including a supply reservoir fed from the tractor emergency glad hand. The trailer service brakes are actuated from the trailer air supply by air relays that get their signal from the tractor service glad hand. The trailer air relays are located adjacent to the trailer front wheels. The distance from the rear of the tow tractor to first set of wheels on the trailer is approximately 10.00 feet, and from the first set of wheels to the second set of wheels is approximately 30.00 feet. The trailer braking system working pressure is 105±5 psi. The fully loaded trailer has a maximum deceleration rate not exceeding 20 ft/s² at 90 psi. Prior tow tractor and munitions trailer testing yielded an actual deceleration rate of approximately 17~19 ft/s² for the MHU-196/204 munitions trailer. The MHU-196 munitions trailer is towed at 10 mi/h loaded and 20 mi/h unloaded. The MHU-204 munitions trailer is towed at 10 mi/h loaded or unloaded. The maximum weight of the fully loaded MHU-196/204 munitions trailer is 80,000 pounds.

7.7 Key words.

Aircraft parking
Flight line
MB-4 tractor
Tow tractor
Tug
Munitions

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CONCLUDING MATERIAL

Custodian:
Air Force – 184

Preparing activity:
Air Force – 184

Agent:
Air Force – 110

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